

EDUCATION

HEC Lausanne

Lausanne, Switzerland

M.Sc. in Quantitative Finance, GPA: 4.95/6.0

2019 – 2021

- **Coursework:** Econometrics; Mathematics; Programming (Python, C++); Statistics
- **Master Thesis:** Rough Volatility Modeling in option pricing validated with a grade of 6.0/6.0

HEC Lausanne

Lausanne, Switzerland

B.Sc. in Economics, GPA: 4.7/6.0

2016 – 2019

- **Coursework:** Econometrics; Mathematics; Programming (Matlab); Statistics

WORK EXPERIENCE

Gartner

Barcelona, Spain

Software Engineer

January 2024 –

- Developed a high-performance API supporting over 100 million users annually across platforms including Capterra, Software Advice, Upcity, and GetApp, implementing the Query Specification pattern to enhance repository flexibility and maintainability while expanding GraphQL schema with new types/resolvers for critical functionalities like segment-based review displays (by category, industry, ...). Designed versatile data models enabling efficient storage across RDBMS (Oracle, PostgreSQL), Elasticsearch, and in-memory systems to optimize performance and scalability
- Redesigned and rebuilt the ad monetization interface as part of an initiative to consolidate fragmented data sources and resolve system inconsistencies that previously disrupted ad delivery. This overhaul streamlined data flow, enhanced front-end reliability, and now supports over XXX million USD in annual revenue.
- Streamlined data ingestion processes by implementing Kafka consumers that read from multiple Confluent topics, using Avro for schema-based deserialization.
- Optimized AWS EKS deployments with Helm and reduced deployment times by 25% by leveraging Amazon ECR remote caching for Docker images, instead of GitHub Actions cache, enabling efficient image layer reuse.
- Spearheaded development of a global query complexity framework with customizable rulesets, enabling back-end teams to safely test algorithm changes in a pre-rollout validation environment and preventing client-side breaks through proactive failure simulation while accelerating iteration cycles by more than 100% on assessment improvements, as the complexity algorithm was only reviewed a few times a year in the past.

Amazon

Iasi, Romania

Software Developer

May 2022 – October 2023

- Developed a high-performance ECS REST API (Flask) integrated with a Bootstrap 5 GUI to automate ticket-raising rules based on OLAP Redshift Database attributes (e.g., seller and brand IDs). This automation reduced staff requirements by 26%, generating over \$1M in yearly savings.

PERSONAL PROJECTS

- **LiveDisplaced:** Developed a data visualization platform using Quart (an asynchronous Flask framework) and SQLAlchemy ORM, integrated with multiple AWS services to track and display the global influx of refugees, asylum seekers, and people of concern. The project aims to raise public awareness by providing accessible up-to-date information through the website <https://www.livedisplaced.com>
- **Async-sendgrid:** An asynchronous client to send email using the famous Twilio service. This client was downloaded more than 300,000 times. <https://pypi.org/project/sendgrid-async/>

LANGUAGES

Native: French, Arabic

Full proficiency: English (IELTS C1)

Partial proficiency: Spanish (B1)

SKILLS

Languages: Python, Elixir, Typescript, Javascript, Java

Technologies: Terraform, Git, Postgres, Mysql, Elasticsearch, AWS(ECS, EKS, Lambda, ECR, IAM, Route 53, ACM, Batch, Cloudfront, Cloudwatch, Aurora, RDS, Secret Manager, ALB, SSM, S3, Fargate), Apache Kafka, Docker, Kubernetes, Gitlab CI, GitHub action, Datadog, Opentelemetry, Confluent